
Product Requirements Document (PRD)

Product: Decentralized Trust Ledger for Academic Representation & Decision Legitimacy

1. Document Metadata

Field	Details
Product Name	Decentralized Trust Ledger for Academic Decision Legitimacy
PRD Version	v1.0
Author(s)	Product Management Team
Reviewers	System Architect, Security Lead, Institutional Advisor
Stakeholders	University Administration, Faculty Committees, Student Bodies
Approval Status	Draft – Ready for Review
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Confidentiality Level	Internal / Academic / Hackathon Submission

2. Executive Summary (1-Page Read)

This product addresses the **loss of trust and legitimacy in internal university decisions** such as class representation, committee formation, and collective approvals. While these processes are frequent and foundational, they are currently managed through informal, centralized, and mutable systems that fail to preserve participation context, mandate strength, or historical accountability.

The solution is a **decentralized decision legitimacy ledger** that cryptographically records *who participated, under what context, and with what representational strength*, without entering political, governmental, or regulatory domains. Blockchain is used strictly as an **immutable trust layer**, not as a voting replacement.

Target users include students, faculty coordinators, and institutional administrators. Success is defined by increased trust in outcomes, verifiable historical decision trails, and institution-wide adoption without legal or administrative friction.

3. Background & Context

Market / Ecosystem Context

Academic institutions increasingly rely on digital tools for governance but lack **trust-preserving infrastructure**. Decisions are often challenged informally, leading to disengagement and legitimacy erosion.

Why This Problem Exists Now

- Institutions are scaling (more students, clubs, hostels, programs)
- Decisions have longer downstream impact
- Digital records exist but are not trustworthy over time

Current Alternatives & Gaps

Alternative	Gap
Paper ballots	No auditability, easily lost
Google Forms	Editable, no identity integrity
Campus ERP modules	Centralized, admin-controlled history
Generic blockchain voting	Over-engineered, political, legally risky

Regulatory / Institutional Context

- Non-governmental
- Institution-scoped
- No statutory or electoral processes involved

4. Problem Statement

Primary Problem

Universities lack a **tamper-proof, verifiable system** to preserve the legitimacy of internal academic decisions over time.

Secondary Problems

- Participation quality is not measurable
- Mandates cannot be audited historically
- Minority dissent disappears
- Decisions lose credibility after leadership changes

Who Experiences the Problem

- Students questioning representation
- Faculty validating committee mandates
- Administrators resolving disputes

When & Where

Occurs during and after:

- CR selection
- Committee formation
- Policy feedback approvals
- Resource prioritization

Impact If Unsolved

- Declining participation
- Informal power concentration
- Repetitive disputes
- Institutional trust erosion

5. Goals & Non-Goals

Goals

Business / Institutional Goals

- Establish trust without legal exposure
- Reduce governance disputes
- Enable historical auditability

User Goals

- Confidence in outcomes
- Transparent participation
- Persistent mandate records

Technical Goals

- Immutable decision records
- Privacy-preserving identity binding
- Campus-scale performance

Non-Goals

- National or political elections
 - Enforcement or disciplinary action
 - Financial governance or tokenization
 - Public or anonymous voting systems
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6. User Personas

Persona 1: Student Participant

- Role: Undergraduate / Postgraduate Student
- Pain Points: Distrust in outcomes, low voice visibility
- Motivation: Fair representation
- Technical Proficiency: Medium
- Success Criteria: Verifiable participation acknowledgment

Persona 2: Faculty Coordinator

- Role: Committee / Class In-charge
- Pain Points: Disputes, legitimacy challenges
- Motivation: Smooth governance
- Technical Proficiency: Medium-High
- Success Criteria: Audit-ready decisions

Persona 3: Institutional Administrator

- Role: Academic Office / Dean's Cell
 - Pain Points: Conflict resolution, compliance concerns
 - Motivation: Stability, scalability
 - Technical Proficiency: High
 - Success Criteria: Low-risk, defensible records
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7. User Journeys & Use Cases

Primary Journey

1. Decision initiated by authorized coordinator

2. Eligible participants notified
3. Participation recorded
4. Decision finalized
5. Legitimacy ledger generated

Secondary Journeys

- Abstention with context
- Minority dissent logging
- Historical mandate lookup

Edge Cases

- Low participation thresholds
- Identity mismatch
- Network interruption

Preconditions

- Institutional identity verification
- Decision scope defined

Postconditions

- Immutable decision record
- Legitimacy metrics available

8. Functional Requirements

Feature 1: Decision Creation Module

- User Story: As a coordinator, I want to define a decision with scope and participants.
- Acceptance Criteria:
 - Decision metadata locked after launch
 - Participant list validated
- Priority: P0

Feature 2: Participation Recording

- User Story: As a student, I want my participation to be recorded immutably.

- Acceptance Criteria:
 - One-person-one-record
 - Timestamped proof
- Priority: P0

Feature 3: Legitimacy Scoring Engine

- Description: Computes participation %, representation spread, dissent markers
- Priority: P1

Feature 4: Historical Ledger View

- Read-only audit trail
- Priority: P1

Feature 5: Privacy Layer

- Vote content off-chain
- On-chain hashes only
- Priority: P0

9. Non-Functional Requirements (NFRs)

Category	Requirement
Performance	≤2s response for campus scale
Scalability	50k users per institution
Availability	99.5%
Reliability	No data loss
Security	End-to-end encryption
Privacy	No public identity exposure
Compliance	GDPR-aligned assumptions
Accessibility	WCAG 2.1 AA
Maintainability	Modular services
Observability	Full audit logs

10. Data Requirements

- Inputs: Identity token, participation action, timestamps
 - Outputs: Legitimacy score, immutable hash
 - Storage: Hybrid (on-chain + off-chain)
 - Retention: Institution-defined
 - Ownership: Institution
 - Auditability: Cryptographic proofs
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11. System Architecture (Conceptual)

- Frontend: Web & mobile interfaces
 - Backend: Decision orchestration service
 - Identity Layer: Institutional authentication
 - Blockchain Layer: Permissioned ledger
 - Analytics: Legitimacy computation engine
 - Trust Boundary: Between UI and ledger
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12. Technical Constraints & Assumptions

- Assumes reliable institutional identity system
 - Permissioned blockchain preferred
 - No public chain dependency
 - Internet access available during participation windows
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13. Security, Privacy & Compliance

- Threat Model: Identity spoofing, data tampering
 - Authentication: Institution-scoped SSO
 - Authorization: Role-based
 - Encryption: AES-256 (data), TLS 1.3 (transit)
 - Compliance: GDPR-inspired principles (assumed)
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14. Risks & Mitigation

Risk	Mitigation
Low adoption	Non-punitive design
Admin resistance	Read-only oversight
Privacy concerns	Off-chain content
Overengineering	Modular rollout
Misinterpretation	Clear non-political framing

15. Metrics & Success Criteria

- North Star: % of decisions with verifiable legitimacy
- Leading: Participation rate
- Lagging: Dispute reduction
- Adoption: Active institutions
- Retention: Repeat usage
- Reliability: Zero integrity failures

16. Rollout & Launch Plan

1. Single class pilot
2. Department-level rollout
3. Campus-wide enablement
4. Optional multi-campus federation

Rollback: Immediate revert to legacy process with data preserved.

17. Open Questions & Future Considerations

- Standard legitimacy thresholds?
- Cross-institution interoperability?
- Long-term archival policies?
- Optional AI-assisted insights (future)

18. Appendix

Glossary

- Decision Legitimacy: Verifiable representational validity
- Ledger: Immutable record store

Explicit Assumptions

- Institution consent obtained
 - No statutory governance involved
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